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# Install missing dependency required by tensorflow-datasets
!pip install -q importlib-resources

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# Experiment 7 : Dog Breed Classification using Transfer Learning
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import matplotlib.pyplot as plt
import tensorflow as tf
import tensorflow_datasets as tfds

# --- 1. Load Dataset (Stanford Dogs) ---
# Srushti Kale - 23070521067: Loading a small subset (10%) to prevent Colab memory/timeout issues
print("Loading Stanford Dogs dataset... (This may take a moment)")
(ds_train, ds_test), ds_info = tfds.load(
    'stanford_dogs',
    split=['train[:10%]', 'test[:10%]'],
    with_info=True,
    as_supervised=True
)

NUM_CLASSES = ds_info.features['label'].num_classes
IMG_SIZE = 224 # MobileNetV2 standard input size

# --- 2. Data Preprocessing (Robust) ---
# Srushti Kale - 23070521067: Resizing and ensuring all images have exactly 3 (RGB) channels
def preprocess_image(image, label):
    # Resize the image first
    image = tf.image.resize(image, (IMG_SIZE, IMG_SIZE))

    # FIX: Check static shape. If it has 1 channel (grayscale), convert it to 3 channels.
    if image.shape[-1] == 1:
        image = tf.image.grayscale_to_rgb(image)
    elif image.shape[-1] is None:
        # Fallback dynamic handling: if shape is completely dynamic, broadcast/tile manually
        channels = tf.shape(image)[-1]
        image = tf.cond(
            tf.equal(channels, 1),
            lambda: tf.image.grayscale_to_rgb(image),
            lambda: image
        )

    # Ensure shape is statically known for the model
    image = tf.ensure_shape(image, (IMG_SIZE, IMG_SIZE, 3))

    # Apply MobileNetV2 specific preprocessing (scales pixels to [-1, 1])
    image = tf.keras.applications.mobilenet_v2.preprocess_input(image)
    return image, label

BATCH_SIZE = 32
train_dataset = ds_train.map(preprocess_image).batch(BATCH_SIZE).prefetch(tf.data.AUTOTUNE)
test_dataset = ds_test.map(preprocess_image).batch(BATCH_SIZE).prefetch(tf.data.AUTOTUNE)

# --- 3. Load Pretrained Model (MobileNetV2) ---
# Srushti Kale - 23070521067: Loading base model without the top layer
base_model = tf.keras.applications.MobileNetV2(
    input_shape=(IMG_SIZE, IMG_SIZE, 3),
    include_top=False,
    weights='imagenet'
)

# Freeze the base model weights
base_model.trainable = False

# --- 4. Build Custom Classification Head ---
# Srushti Kale - 23070521067: Adding Global Average Pooling and Dense layers for breeds
model = tf.keras.Sequential([
    base_model,
    tf.keras.layers.GlobalAveragePooling2D(),
    tf.keras.layers.Dropout(0.2),
    tf.keras.layers.Dense(NUM_CLASSES, activation='softmax', name='predictions')
])

model.summary()

# --- 5. Compile and Train ---
# Srushti Kale - 23070521067: Compiling the model
model.compile(
    optimizer=tf.keras.optimizers.Adam(learning_rate=0.001)

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optimizer=tf.keras.optimizers.Adam(learning_rate=0.001),
loss='sparse_categorical_crossentropy',
metrics=['accuracy'])

)

print("\nTraining Custom Head on Stanford Dogs...")
history = model.fit(
    train_dataset,
    epochs=5,
    validation_data=test_dataset
)

# --- 6. Evaluate and Visualize ---
loss, accuracy = model.evaluate(test_dataset)
print(f"\n--- Evaluation Results (Srushti Kale - 23070521067) ---")
print(f"Test Accuracy: {accuracy * 100:.2f}%")

plt.figure(figsize=(8, 5))
plt.plot(history.history['accuracy'], label='Training Accuracy')
plt.plot(history.history['val_accuracy'], label='Validation Accuracy')
plt.title('Transfer Learning Accuracy (MobileNetV2) - Srushti Kale (23070521067)')
plt.xlabel('Epoch')
plt.ylabel('Accuracy')
plt.legend()
plt.grid(True)
plt.show()
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Loading Stanford Dogs dataset... (This may take a moment)
Downloading data from https://storage.googleapis.com/tensorflow/keras-applications/mobilenet_v2/mobilenet_v2_weights_tf_dim_9406464/9406464 0s 0us/step
Model: "sequential"

Layer (type)	Output Shape	Param #
mobilenetv2_1.00_224 (Functional)	(None, 7, 7, 1280)	2,257,984
global_average_pooling2d (GlobalAveragePooling2D)	(None, 1280)	0
dropout (Dropout)	(None, 1280)	0
predictions (Dense)	(None, 120)	153,720

Total params: 2,411,704 (9.20 MB)
Trainable params: 153,720 (600.47 KB)
Non-trainable params: 2,257,984 (8.61 MB)

Training Custom Head on Stanford Dogs...
Epoch 1/5
38/38 106s 3s/step - accuracy: 0.1600 - loss: 3.9936 - val_accuracy: 0.4394 - val_loss: 2.6397
Epoch 2/5
38/38 92s 2s/step - accuracy: 0.7508 - loss: 1.3855 - val_accuracy: 0.6142 - val_loss: 1.6235
Epoch 3/5